HARDWARE STUDY / PRODUCT ARCHITECTURE

Reverse Engineering of Samsung Galaxy A30s

From physical inspection to product architecture: what is inside, and why it was designed that way.

Galaxy A30s: Hardware at a Glance

PROCESSOR

Exynos 7904

14 nm, 2 × A73 + 6 × A53

GPU + MODEM

Mali-G71 MP2

LTE Cat.12 on-chip

MEMORY

4 GB LPDDR4

K4FBE3D4HM

STORAGE

64 GB V-NAND

KLMCG2UCTA

DISPLAY

6.4-inch Super AMOLED

Optical in-display fingerprint

BATTERY

4000 mAh Li-Po

Held by double-sided tape

POWER ICs

S2MPU06B

PMIC, plus ET9553M charger

PORTS

USB-C + 3.5 mm

Nano-SIM / microSD shared tray

158.5 × 74.7 × 7.8 mm | Sensors: accelerometer, gyroscope, proximity, ambient light, magnetometer

From Physical Evidence to Safe Teardown

1. Inspect before opening

- Labels: IMEI, serial, model, BIS sticker, battery label
- 16 screws: 14 on the mid-frame, 1 mainboard, 1 sub-board
- Seams: back cover, display, SIM tray, ports, speaker grille

2. Follow the process

- Document, photograph, open, photograph, identify, record, analyze
- Tools: precision drivers, pry tools, spudger, tweezers, multimeter
- Keep screws in labeled containers

3. Stay safe

- ESD wrist strap and mat; handle PCB edges only
- Disconnect power before disassembly
- Batteries and capacitors retain charge, so never dismantle energized parts casually

From Components to System Architecture

01
Observation
Connectors, ICs and a battery

02
Identification
This IC is a specific processor

03
Specification
Datasheet: voltage, peripherals

04
Function
It drives sensors and display

05
Reasoning
Why this part and this architecture?

Battery → power management (S2MPU06B, ET9553M) → Exynos 7904 → memory, display, sensors and radios

Reverse engineering reveals not only what is inside a product, but why it was designed that way.

